How we chose the variables

1. **Rank genres by total streams.** Hip-Hop/Rap (1,224 tracks), Pop (1,633) and Latin (342). Other/Unknown is excluded. [Add total streams.]
2. **Review correlations.** Pooled across the three genres, every correlation with log streams is weak ($|r| \leq 0.17$; release year +0.17, duration −0.12, log followers +0.05, liveness +0.03). [Add genre-focused matrices.]
3. **Fit one candidate model per genre.** The same 19 variables in each genre; read every estimate and p-value.
4. **Select focus variables after that comparison.** Rule: $p < 0.05$ in the genre's candidate model and a clear business meaning.
5. **Build and evaluate.** Parallel vs. interaction models, then a genre-specific final model: adjusted R^2, RMSE, p-values, Q-Q and residual plots. Then predict with all other predictors held constant.

STEP 3 · CANDIDATE MODEL FIT (19 VARIABLES)

Genre	n	Overall p	Adj. R^2
Hip-Hop/Rap	1,224	2.2e-13	0.067
Pop	1,633	< 2.2e-16	0.061
Latin	342	5.2e-9	0.165

STEP 4 · FOCUS VARIABLES SELECTED

Hip-Hop/Rap: log followers (p = 2.8e-5)

Pop: release size (2–6 tracks p = 2.0e-4; 7+ tracks p = 2.1e-5)

Latin: duration (p = 0.0014) and liveness (p = 0.038)

How to read the coefficients: the outcome is ln(streams). A coefficient is the change in log streams for a one-unit increase in that variable (or for a category vs. its reference), holding the other predictors constant. Predicted streams come from predict() on the model.

Pop: release size findings

PARALLEL VS. INTERACTION

Model (pooled)	Adj. R^2
Parallel	0.0770
Interaction	0.0812

ANOVA F = 4.66, p = 0.0009: interaction wins.

Parallel: vs. a single, a 2–6 track release changes log streams by −0.321 and a 7+ track release by −0.160, others held constant.

Slope change: Pop differs from Hip-Hop/Rap by −0.553 (2–6) and −0.246 (7+), for net Pop effects of −0.558 and −0.266.

FINAL POP MODEL (n = 1,633)

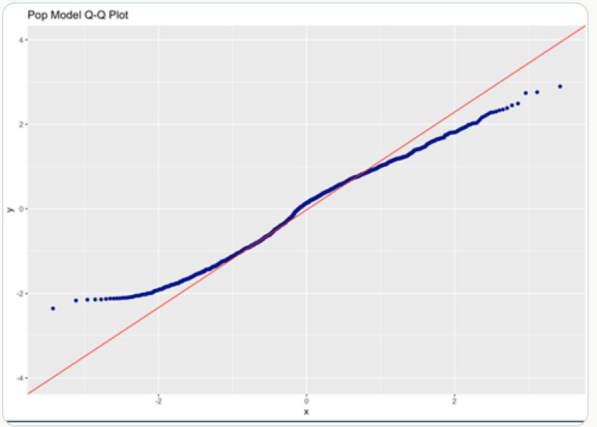
log(streams) = β0 + β1 release size + β2 explicit + β3 release year + β4 log(followers) + β5 duration

Coefficients: vs. a single, 2–6 tracks change log streams by −0.568 and 7+ tracks by −0.274 (both p < 0.001), others held constant.

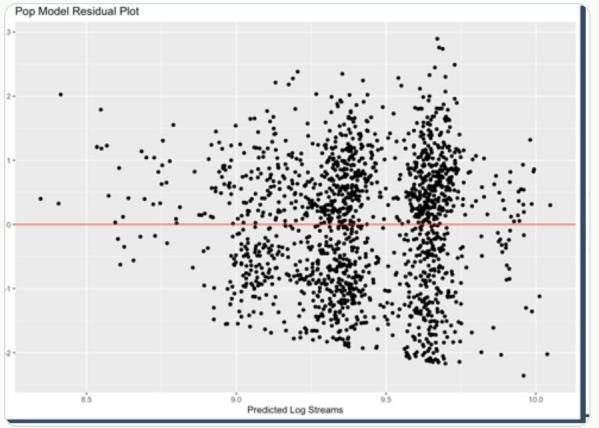
Adjusted R^2 = 0.036: the model explains 3.6% of the variability in Pop log streams.

Overall p = 4.3e-12 · RMSE = 1.28

Q-Q plot



Residuals vs. predicted log streams



Reliability: the residual plot shows no curve or funnel (the stripes are the release-size groups). The Q-Q points leave the line at both ends in an S-shape, so p-values are approximate. Read direction and size, not exact p-values.

Source: summary(model_release_parallel), summary(model_release_interaction), anova(...), summary(model_pop), pop_points plots.

Pop: business recommendation

Lead with singles; the single's advantage appears to be growing.

PREDICTED STREAMS, OTHERS HELD CONSTANT: NON-EXPLICIT · 2017 · 4.4M FOLLOWERS · 3.5 MIN

Release	Predicted streams	Difference vs. single
1 track	15,220	baseline
2–6 tracks	8,628	−6,592
7+ tracks	11,574	−3,646

Reading the coefficients: vs. a single, a 2–6 track release lowers log streams by 0.568 and a 7+ track release by 0.274.

Over time (release size × year): a single's log streams rise by 0.218 per year. The slope changes by −0.229 for 2–6 tracks and −0.204 for 7+ tracks, so those releases net −0.011 and +0.014 per year (interaction terms $p < 0.001$). Adjusted R^2 rises from 0.0356 to 0.0609.

RECOMMENDATION

Prioritize singles over 2–6 or 7+ track releases in Pop. The single's slope is rising while the others are about flat, so its edge appears to be widening.

NEXT STEP

Map actual Pop streams by release size and year, and run the ANOVA for release size × year (not yet run).

Associations, not causal effects. The model explains 3.6% of the variability.

Source: pop_predictions; summary(model_pop_time_interaction); summary(model_pop).

Latin: duration and liveness findings

PARALLEL VS. INTERACTION

Model (pooled)	Adj. R^2
Parallel	0.0775
Interaction	0.0891

ANOVA F = 11.19, p = 5.1e-9: interaction wins.

Parallel: each extra minute changes log streams by −0.159 (p < 0.001); each one-unit increase in liveness by +0.269 (p = 0.105, not significant).

Slope change: Latin differs from Hip-Hop/Rap by −0.389 per minute (net −0.486) and +1.804 per liveness unit (net +1.834). Pop's changes are not significant.

FINAL LATIN MODEL (n = 342)

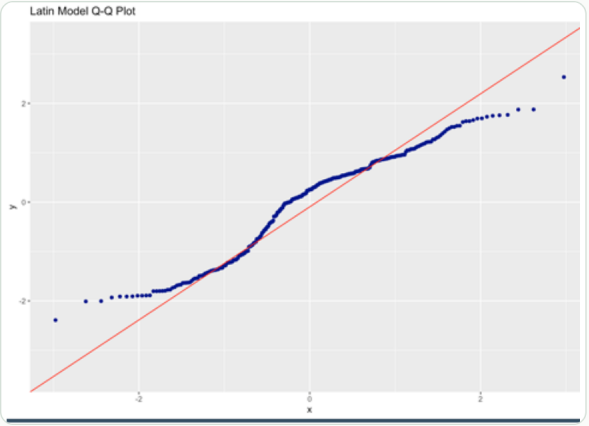
log(streams) = β0 + β1 duration + β2 liveness + β3 explicit + β4 release size + β5 release year + β6 log(followers)

Coefficients: each extra minute changes log streams by −0.405 (p = 0.0001); each one-unit increase in liveness by +1.490 (p = 0.002), others held constant.

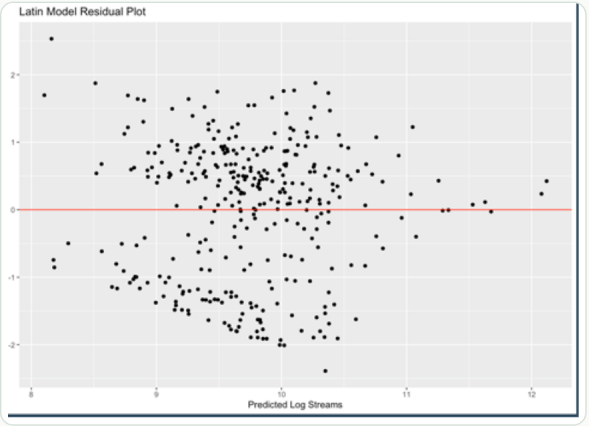
Adjusted R^2 = 0.126: the model explains 12.6% of the variability in Latin log streams.

Overall p = 5.3e-9 · RMSE = 1.50

Q-Q plot



Residuals vs. predicted log streams



Reliability: no clear curve or funnel in the residual plot, but the Q-Q plot is S-shaped and there are only 342 tracks, so p-values are approximate. Read direction only.

Source: summary(model_tracks_parallel), summary(model_tracks_interaction), anova(...), summary(model_latin), latin_points plots.

Latin: business recommendation

Shorter, livelier tracks go with more streams.

PREDICTED STREAMS, OTHERS HELD CONSTANT: NON-EXPLICIT · SINGLE · 2017 · 9.7M FOLLOWERS

Scenario	Streams	Difference
3 min, liveness 0.128	29,625	baseline
4 min, liveness 0.128	19,764	−9,861
3.5 min, liveness 0.10	23,113	baseline
3.5 min, liveness 0.50	41,952	+18,840*

*Liveness 0.50 is far above the Latin median (0.128), so treat this row as a stretch of the data.

Reading the coefficients: each extra minute lowers log streams by 0.405; each one-unit increase in liveness raises log streams by 1.490.

Slope change vs. Hip-Hop/Rap: −0.389 per minute and +1.804 per liveness unit (both $p < 0.001$).

RECOMMENDATION

Favor shorter Latin tracks and pilot live-sounding versions, then measure the result. Treat both as hypotheses to test, not guarantees.

NEXT STEP

Split Latin by subgenre to see what liveness captures, and re-check with more tracks (n = 342).

Associations, not causal effects. The model explains 12.6% of the variability.

Source: latin_predictions; summary(model_latin); summary(model_tracks_interaction).

Hip-Hop/Rap: artist followers findings

PARALLEL VS. INTERACTION

Model (pooled)	Adj. R^2
Parallel	[add]
Interaction	[add]

ANOVA: [add F and p]. Not yet run (section 10 of the script).

Separate-genre slopes per one-unit increase in log followers: Hip-Hop/Rap +0.107 ($p < 0.001$), Pop +0.005 ($p = 0.76$), Latin +0.051 ($p = 0.39$).

Slope change: [add after the interaction model is run].

FINAL HIP-HOP/RAP MODEL (n = 1,224)

log(streams) = β0 + β1 log(followers) + β2 explicit + β3 release size + β4 release year + β5 duration

Coefficient: each one-unit increase in log followers (followers × 2.7) changes log streams by +0.107 ($p = 8.0e-8$), others held constant.

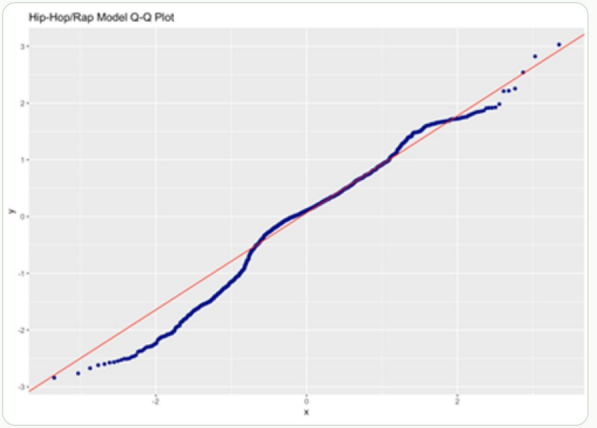
Adjusted R^2 = 0.057: the model explains 5.7% of the variability in Hip-Hop/Rap log streams.

Overall p = 1.1e-14 · RMSE = 1.27

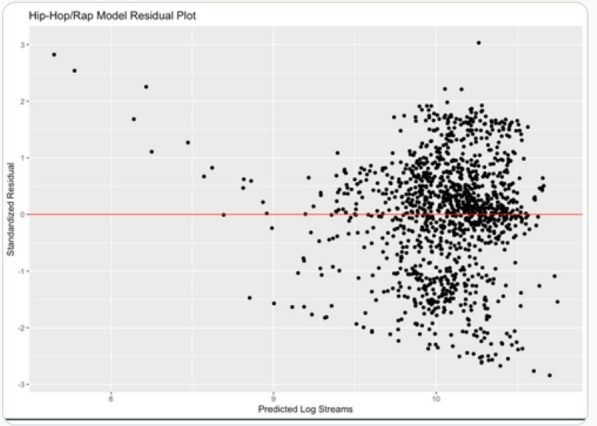
Reliability: even spread for most predictions, with a line of points sloping down at low predicted values. The Q-Q points leave the line at both ends in an S-shape, so p-values are approximate. Read direction and size only.

Source: summary(model_hiphop), hiphop_points plots, summary(model_pop), summary(model_latin). Followers parallel vs. interaction: not run.

Q-Q plot



Residuals vs. predicted log streams



Hip-Hop/Rap: business recommendation

Audience size gives a modest lift, and it is not detectable in Pop or Latin.

PREDICTED STREAMS, OTHERS HELD CONSTANT: NON-EXPLICIT · SINGLE · 2018 · 3.4 MIN

Artist followers	Streams	Difference
100,000	15,669	baseline
1,000,000	20,043	+4,374

Reading the coefficient: each one-unit increase in log followers raises log streams by 0.107. Going from 100,000 to 1,000,000 followers is an increase of 2.30 log units.

Other genres, same coefficient: Pop +0.005 (p = 0.76) and Latin +0.051 (p = 0.39) are not significant. The pooled correlation of log followers with log streams is 0.05.

Slope change by genre: [add once the followers interaction model is run].

RECOMMENDATION

In Hip-Hop/Rap, an established audience helps modestly. In Pop and Latin, artist size is not a detectable driver, which suggests room for up-and-coming artists. Confirm with the followers interaction test.

NEXT STEPS

Run followers parallel vs. interaction. Test followers × year to see whether artist size matters more or less over time.

Associations, not causal effects. The model explains 5.7% of the variability.

Source: hiphop_predictions; summary(model_hiphop), summary(model_pop), summary(model_latin).